

MethylCentroid Theoretical Foundation

MethylPipeline

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1 Goal

MethylCentroid summarises a cohort of methylation samples without storing the full sample $\times$ position matrix in memory.

The centroid stores per-position *sufficient statistics* plus an empirical histogram so that downstream comparison can remain ECDF-based.

2 Per-Sample Quantities

For sample j at genomic position i , define

$$mC_{ij} \quad \text{methylated count,} \tag{1}$$

$$uC_{ij} \quad \text{unmethylated count,} \tag{2}$$

$$c_{ij} = mC_{ij} + uC_{ij} \quad (\text{coverage}), \tag{3}$$

$$x_{ij} = \frac{mC_{ij}}{c_{ij}} \in [0, 1] \quad (\text{methylation fraction, defined when } c_{ij} > 0). \tag{4}$$

Only samples with $c_{ij} > 0$ contribute to position i 's centroid statistics.

3 Stored Sufficient Statistics

For each genomic position the centroid stores the seven accumulators shown in Table 1.

Symbol	Code name	Definition
N	N	$\sum_j 1$
S_x	Sx	$\sum_j x_{ij}$
S_{x^2}	Sx2	$\sum_j x_{ij}^2$
Σm	Sm	$\sum_j mC_{ij}$
Σu	Su	$\sum_j uC_{ij}$
Σc^2	Sc2	$\sum_j c_{ij}^2$
Σwx^2	Swx2	$\sum_j c_{ij} x_{ij}^2$

Table 1: Sufficient statistics stored per genomic position.

3.1 Derived Quantities

The stored accumulators are enough to recover commonly needed summaries without re-reading sample files.

Mean methylation fraction.

$$\hat{\mu} = \frac{S_x}{N}. \quad (5)$$

Sample variance of methylation fractions.

$$\hat{\sigma}^2 = \frac{S_{x^2} - S_x^2/N}{\max(N-1, 1)}. \quad (6)$$

Average counts.

$$\widehat{mC} = \frac{\Sigma m}{N}, \quad \widehat{uC} = \frac{\Sigma u}{N}. \quad (7)$$

Method-of-moments Beta parameters (derived, not stored). When a downstream consumer requests Beta parameters, they are computed from $\hat{\mu}$ and $\hat{\sigma}^2$:

$$\hat{\alpha} = \hat{\mu} \left(\frac{\hat{\mu}(1 - \hat{\mu})}{\hat{\sigma}^2} - 1 \right), \quad (8)$$

$$\hat{\beta} = (1 - \hat{\mu}) \left(\frac{\hat{\mu}(1 - \hat{\mu})}{\hat{\sigma}^2} - 1 \right). \quad (9)$$

These parameters are *not* part of the comparison pipeline; they are available only as derived statistics.

Heterogeneity statistics. Sc2 and Swx2 allow coverage-weighted variance decompositions, for example DerSimonian–Laird-style between-sample heterogeneity estimates used by the `dl_heterogeneity` helper in MethylUtils. This is why both fields remain part of the persisted schema even though they are not used in the core centroid mean or variance formulas.

4 ECDF Histogram

Each supported centroid must also store a per-position histogram over methylation fractions:

- **bins** — number of histogram bins $B \geq 1$.
- **bin_counts** — array of shape $(n_{\text{positions}}, B)$.
- **bin_edges** — reconstructed in memory as uniform edges on $[0, 1]$: $0, 1/B, 2/B, \dots, 1$.

The parameter `binned_stats.bins` is **mandatory** and must satisfy $B \geq 1$. The canonical default is $B = 20$.

For each accumulator update with sample j , the bin index is

$$b_{ij} = \lfloor x_{ij} \cdot B \rfloor \text{ clamped to } \{0, \dots, B-1\}, \quad (10)$$

and `bin_counts` $[i, b_{ij}]$ is incremented.

The histogram defines an *empirical CDF* at the bin edges. A PCHIP (Piecwise Cubic Hermite Interpolating Polynomial) spline is used to reconstruct a continuous CDF $F(x)$ and PDF $f(x) = F'(x)$ for any $x \in [0, 1]$ when downstream code evaluates overlap or log-likelihood.

5 Why ECDF Only

The current contract removes runtime switching among Normal, Beta, Beta-Binomial, and Beta-Mixture comparison modes. The ECDF-only design offers several advantages:

- **Shape preservation.** The empirical histogram captures skewed, multimodal, or otherwise irregular methylation distributions without imposing a parametric family.
- **Contract alignment.** Build-time (MethylCentroid) and compare-time (MethylCentroid-Pair, MethylDetector) both rely on the same data model: sufficient statistics plus empirical histograms.
- **Downstream consistency.** ECDFClassifier (MethylDetector classifier stage) uses the same PCHIP PDF model as the DMP detection stage, so the score that selects DMPs is consistent with the score that classifies new samples.

6 Cohort Update Semantics

At the runner level, cohort membership is resolved as

$$\text{effective_samples} = (\text{samples} \setminus \text{remove_samples}) \cup \text{add_samples}. \quad (11)$$

That resolved cohort is built for the target chromosome/context and persisted. The final active cohort is written to:

- HDF5 metadata field `samples_used`,
- sidecar config field `samples` in `{chrom}-{ctx}_config.json`.

7 HDF5 Schema

The centroid file stores only `methylation_data` datasets; no separate `binned_stats` group is used.

Dataset / Attribute	Content
<code>methylation_data/pos</code>	Genomic positions (uint32)
<code>methylation_data/tnc</code>	Trinucleotide context (uint8)
<code>methylation_data/N</code>	Sample count per position
<code>methylation_data/Sx</code>	Sum of methylation fractions
<code>methylation_data/Sx2</code>	Sum of squared fractions
<code>methylation_data/Sm</code>	Sum of methylated counts
<code>methylation_data/Su</code>	Sum of unmethylated counts
<code>methylation_data/Sc2</code>	Sum of squared coverages
<code>methylation_data/Swx2</code>	Coverage-weighted sum of squared fractions
<code>methylation_data/bin_counts</code>	ECDF histogram ($n_{\text{pos}} \times B$)
<code>methylation_data.attrs[bins]</code>	Scalar bin count B

Bin edges are *not* stored; they are derived as uniform edges on $[0, 1]$ from the stored bin count B .

8 Summary

`MethylCentroid` is built around four design principles:

1. Compact sufficient statistics $(N, S_x, S_{x^2}, \Sigma m, \Sigma u, \Sigma c^2, \Sigma wx^2)$.
2. Required ECDF histogram data (`bin_counts`, $B \geq 1$).
3. Deterministic cohort updates via set arithmetic.
4. One downstream comparison mode: ECDF.